Cartier–Miller evaluation of genus-one coefficient sums

David Jordan

E-mail: David.jordan@mu.ie

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Abstract

Let $p \geq 7$ be prime and let $f \in \mathbb{F}_p[w]$ be squarefree of degree three or four with $f(0) = 1$. We express every admissible base-field value of the truncation $T_f = f^{(p-1)/2} \bmod w^p$ through the elliptic trace and a normalized logarithmic derivative. The identity includes trace ± 1, with the residue-divisor normalization proved directly in characteristic p for both signs. A differentiated Miller recurrence gives $O(\log p)$ field operations per query after trace preparation, and Schoof's algorithm gives deterministic polynomial-time preparation. A quadratic algebra avoids square-root and nonresidue searches. A corollary evaluates rationally weighted quadratic character sums. The original quarter-point weighted central-binomial sum is a worked application.

Introduction and main theorem

Consider evaluation at a base-field point of the degree-below-p truncation of a polynomial power. Put $h = (p-1)/2$ and write

$$f(w)^h = \sum_m c_m w^m, \qquad T_f(w) = \sum_{m=0}^{p-1} c_m w^m, \qquad H = c_{p-1}. \tag{1}$$

For $\lambda \neq 0$ in $\mathbb{F}_p$, the weighted coefficient expression is exactly this truncation evaluated at $\mu = 1/\lambda$:

$$S_f(\lambda) = [w^{p-1}] \frac{f(w)^h}{1-\lambda w} = \sum_{m=0}^{p-1} c_m \lambda^{p-1-m} = T_f(1/\lambda). \tag{2}$$

The last equality uses $\lambda^{p-1} = 1$ and is a base-field identity. At $\lambda = 0$, the coefficient expression gives $S_f(0) = H$; evaluation $T_f(0) = 1$ is separately immediate. Euler's criterion also gives $T_f(\mu) = \chi - \sum_{m \geq p} c_m \mu^m$, where $\chi = (f(\mu)/p)$ when $f(\mu) \neq 0$.

Let E be the smooth projective curve $v^2 = f(w)$ with origin $O = P_+ = (0,1)$, invariant differential $\omega = dw/v$, and exact trace $\tau = p + 1 - \#E(\mathbb{F}_p)$. We call $\lambda \in \mathbb{F}_p$ admissible if $\lambda = 0$, or if $\lambda \neq 0$ and $f(1/\lambda) \neq 0$. For a nonzero admissible argument define

$$\begin{gathered} d = f(1/\lambda), \quad v_0^2 = d, \quad \chi = (d/p), \\ Q_\pm = (1/\lambda, \pm v_0), \quad D = (Q_+) - (Q_-), \quad r = \tfrac{1}{\lambda v_0}. \end{gathered} \tag{3}$$

For an integer M annihilating the class of D, fix a rational function F_D with $\operatorname{div}(F_D) = MD$, and set $\Psi_M(D) = ((dF_D/F_D)/\omega)(O)$. This value is independent of the scalar multiplying F_D. For $p \nmid M$ it retains the specified divisor representative; for $M = p$ it depends only on its divisor class.

Theorem 1 (Uniform base-field evaluation). Under these hypotheses, $H \equiv \tau \pmod p$, and every nonzero admissible argument satisfies

$$S_f(\lambda) = H - \chi r \Psi_{M_\chi}(D), \qquad M_\chi = p + 1 - \chi\tau. \tag{4}$$

The product $r\Psi_{M_\chi}(D)$ belongs to $\mathbb{F}_p$ and is unchanged by replacing v_0 by $-v_0$. The formula includes $\tau = \chi$, when $M_\chi = p$, and requires no division by that annihilator. Corollary 4 states the evaluation and preparation costs.

The query bound excludes preparation of the exact trace: for one query on a new curve, point counting can dominate the total cost, while repeated queries on the same curve and prime reuse that preparation. The main theorem concerns genus one, exponent $(p-1)/2$, coefficient cutoff $p-1$, values modulo p, non-branch evaluation arguments, and $f(0) \neq 0$ after the stated normalization.

The regular identity follows from cancellation of residues and Cartier semilinearity. Lemma 3 supplies the exceptional normalization by identifying a Cartier defect with the established characteristic-p additive transfer. The contributions considered here are the coefficient-sum identity, its explicit scalar for both signs, the trace -1 residue-divisor formulation, and the algorithmic consequence. The transfer, the differentiated Miller recurrence and the trace 1 identification of the Artin–Schreier quasi-period with a logarithmic differential are established (Gunji, 1976; Voloch, 1990, 1997, n.d.; Semaev, 1998). No priority claim is made for the underlying additive map.

Truncated binomial and hypergeometric congruences motivate the evaluation problem. The question is which large truncations can be evaluated through a small geometric object rather than accumulated term by term. Direct accumulation at a cutoff proportional to p takes $\Theta(p)$ field operations, while polynomial-block recurrence methods have approximately square-root cost up to logarithmic factors (Bostan et al., 2007). The result here fixes the coefficient cutoff at $p-1$ and varies the evaluation point. It does not supply a uniform polylogarithmic algorithm for arbitrary central-binomial stopping indices. The quarter-point application shows how its particular prefix and boundary coefficient recover an original weighted sum. The rounded-third subsection separates another fast prefix from its additional factorial-equivalent boundary and records the scope of the companion implementations.

The implementations address three different workloads: one complete evaluation at a new prime, repeated admissible queries over one prepared curve and field, and a table across many primes. In the rounded-third family the trace is known to be zero, but the weighted sum also requires a boundary coefficient. The integrated evaluator combines its Miller prefix with an exact boundary backend, whereas the dense-table backend acquires both quantities jointly through a shared recurrence. Their costs are compared on matched outputs in the appendix.

The classical Hasse–Witt congruence is recovered at $\lambda = 0$, where $S_f(0) = H \equiv \tau \pmod p$. For nonzero admissible λ, the differential θ_λ has two simple poles. Theorem 1 extends this coefficient selection from a holomorphic differential to that third-kind family. This is a third-kind extension of the Hasse–Witt identity; the terminology does not assert a separate deformation-theoretic theorem.

Corollary 1.1 (Weighted quadratic character sum). Let $\mu \in \mathbb{F}_p^\times$ with $f(\mu) \neq 0$, put $\lambda = \mu^{-1}$ and use the data of Theorem 1. Define $\varepsilon_f(w) = \left(\frac{f(w)}{p}\right)$, including the value zero at a root of f, and put $\chi = \varepsilon_f(\mu)$. Then

$$\sum_{\substack{w \in \mathbb{F}_p \\ w \neq \mu}} \frac{\varepsilon_f(w)}{w - \mu} = \frac{H - \chi}{\mu} + \frac{\chi f'(\mu)}{2f(\mu)} - \frac{\chi}{v_0}\Psi_{M_\chi}(D). \tag{5}$$

Proof. The independent point-sum calculation below gives $S_f(\lambda) = \chi - \chi\mu f'(\mu)/(2f(\mu)) + \mu\sum_{w\neq\mu}\varepsilon_f(w)/(w-\mu)$. Substitute (4) and $r = \mu/v_0$. Corollary 4 gives the same prepared-query cost for this character sum.

A useful multiplicative comparison is the classical finite logarithm, studied in the finite-polylogarithm literature (Elbaz-Vincent & Gangl, 2002):

$$\ell_{1,p}(X) = \sum_{j=1}^{p-1} \frac{X^j}{j} \equiv \frac{1 - X^p - (1-X)^p}{p} \pmod p. \tag{6}$$

The quotient on the right is formed in $\mathbb{Z}[X]$ before reduction. Its coefficients follow from $\binom{p}{j}/p \equiv (-1)^{j-1}/j \pmod p$. At a base-field value, modular powers modulo p^2 and exact division by p evaluate

it with $O(\log p)$ ring operations. Both this example and Theorem 1 replace a long truncation by short group arithmetic, but this comparison does not identify the elliptic expression with a multiplicative Fermat quotient or establish a general theory of elliptic finite polylogarithms.

Relation to prior work

Serre (1958, Propositions 8 and 10) records Cartier's properties and identifies geometric p-torsion divisor classes with Cartier-fixed holomorphic differentials through dlog. Semaev (1998, Lemmas 1–3) uses this additive transfer and a differentiated Miller recurrence for discrete logarithms on elliptic p-torsion. With our invariant differential fixed, his local evaluation $(F_D'/F_D)(R)$ is $\Psi_p(D)$ times the local coefficient of ω at R. Rück (1999) extends the construction to divisor class groups of curves.

For trace 1, the curve has exactly p rational points and is anomalous; for trace -1, its quadratic twist is anomalous. Satoh and Araki (1998, corrected in 1999) use elliptic Fermat quotients, and Smart (1999) uses a p-adic lift, to solve the anomalous-curve discrete-logarithm problem. These belong to the same established setting. Voloch (n.d., pp. 3–4) explicitly relates his descent map to Semaev's logarithmic transfer and compares it with the Smart and Satoh–Araki maps. Thus the identification of these maps is also prior work; the purpose here is to specify the coefficient-sum scalar relative to ω and the actual residue divisor.

In short Weierstrass coordinates, write $F(x)^{(p-1)/2} = \mathcal{U}(x) + Hx^{p-1} + x^p\mathcal{V}(x)$, with $\deg \mathcal{U} \le p-2$. When the trace is 1, $H = 1$ and Verschiebung is zero on $E(\mathbb{F}_p)$, since $\pi + V = 1$ and π is the identity on rational points. Voloch's descent map therefore gives an injective homomorphism $E(\mathbb{F}_p) \to \mathbb{F}_p$, represented by $y\mathcal{V}(x)$ (Voloch, 1990, Proposition 1.3; n.d., p. 3). Once its coefficients have been prepared, evaluating this polynomial by Horner's rule takes $O(p)$ field operations; this statement does not include expanding $F^{(p-1)/2}$. The established logarithmic-transfer recurrence evaluates the same normalized map in $O(\log p)$ operations (Semaev, 1998).

For trace 1 the exceptional normalization is classical in substance. The high-part Artin–Schreier construction goes back to Deuring (1941; see Voloch, 1990, Lemma 1.1, and 1997, Section 2). Cassels (1971, Sections 1, 5–6) writes the cover as $g^p - Hg = y\mathcal{V}(x)$ and relates translations by kernel points to additive quasi-periods. Gunji (1976) studies the logarithmic differentials of functions with divisor $p((P) - (O))$ and their Cartier invariance. Voloch (1990, Lemma 1.1 and Proposition 1.3) identifies the cover with Verschiebung and obtains the descent homomorphism. Voloch (1997, Proposition 1 and the discussion after Proposition 3) develops its quasi-periodicity and its relation to logarithmic derivatives; he also corrects the differential scale and the sign convention of the 1990 construction.

Lemma 3 assembles these ingredients over a finite field. It fixes the scalar relative to $\omega = dw/v$ and the actual residue divisor, and includes trace -1, where a factor 2 enters. We have not located this negative-eigenvalue residue-divisor statement, or the two-pole base-field form of Theorem 1, in the sources considered here; that observation does not establish novelty. Frobenius acting on third-kind differentials also occurs in Coleman–Gross height computations (Balakrishnan & Besser, 2012), and Cartier actions on polynomial-power coefficients appear in Dwork crystals (Beukers & Vlasenko, 2021). These are adjacent frameworks.

There is also a distinct many-primes workload. Harvey and Sutherland (2014) compute Hasse–Witt matrices modulo all good primes below a bound in average polynomial time. Their amortization across primes differs from the present preparation of one curve over one field followed by point-evaluation queries. The rounded-third companion specializes this many-primes approach to a triangular recurrence; its amortized workload is distinct from both a single-prime query and repeated queries over one prepared field.

The rounded-third boundary implementation uses small-conductor Jacobi relations and interval products, following van Wamelen (2002) and the split construction considered by Tal (2026). The relevant part is Tal's split construction with a prime $\ell \mid p-1$, using Jacobi congruences to recover ratios of grid factorials and then short interval corrections. This is distinct from the extension-field construction for divisors of $1+p+p^2$ highlighted in Tal's abstract; no exponent bound from that construction is as-

serted here. The companion specializes the anchor intervals to the rounded-third index and supplies exact reconstruction certificates. The present integration connects that arithmetic to the existing Miller prefix and complete-sum interface.

Notation

Table 1. Principal notation and local conventions.

Symbol	Meaning
f, w, v, O, ω	Normalized cubic or quartic model, coordinates, distinguished origin and fixed differential
T_f, S_f, H, τ	Truncation, coefficient sum, Hasse coefficient and integer trace
$\pi, \sigma, \mathcal{C}$	Point Frobenius, coefficient conjugation and Cartier operator
χ, ε_f	Character at the argument and the quadratic character function
D, M_χ, Ψ_M	Specified two-pole divisor, annihilator and normalized logarithmic derivative
K, Θ_R	Torsion kernel and third-kind differential in Lemma 3
$\mathcal{G}_f, g$	High-part value and a rational function defining a principal change
$a_i, B_L, U_p(L)$	Central-binomial coefficient, strict prefix and weighted sum
m, L, G	Rounded-third indices and $G = m! \bmod p$
ℓ, k_ℓ, N	Prime Jacobi conductor, grid step $(p-1)/\ell$ and denominator-12 anchor index
$\mathcal{T}, \mathcal{S}, \mathcal{D}$	Low and high cubic-power parts and differential operator in Proposition 5
Δ_j, A_j	Cleared factorial denominator and forest recurrence matrix
$N_{\mathrm{qry}}, q_k, e_k$	Number of queries, extension-field size and extension character sign

Cartier evaluation and the regular identity

All differential identities are in characteristic p. Frobenius π on points means the p-power map on coordinates. Coefficient conjugation σ raises constants to their pth powers while leaving coordinate variables fixed. Put

$$\omega = \frac{dw}{v}, \qquad \theta_\lambda = \frac{dw}{(1-\lambda w)\, v}. \tag{7}$$

The differential ω is holomorphic and nowhere zero. For nonzero admissible λ, the only poles of θ_λ are simple poles at Q_+ and Q_- with residues $-r$ and r.

Let $\mathcal{C}$ denote absolute Cartier, and let id denote the identity on rational differentials. We use the local convention

$$\mathcal{C}\left(\sum_n b_n z^n\, dz\right) = \sum_j b_{pj+p-1}^{1/p} z^j\, dz. \tag{8}$$

It satisfies $\mathcal{C}\,(a^p \eta) = a\mathcal{C}(\eta)$ and $\mathcal{C}\,(dF/F) = dF/F$ (Cartier, 1957; Serre, 1958, Proposition 8). Since $f(0) = 1$, w is a uniformiser at O. The relation $v^{p-1} = f^h$ gives the coefficient selection directly:

$$\begin{aligned} \theta_\lambda &= v^{-p} \tfrac{f^h}{1-\lambda w}\, dw, \\ \mathcal{C}(\omega) = H\omega, \qquad & \left(\tfrac{\mathcal{C}(\theta_\lambda)}{\omega}\right)(O) = S_f(\lambda)^{1/p}. \end{aligned} \tag{9}$$

Here $H \in \mathbb{F}_p$, so $H^{1/p} = H$. The trace relation is the genus-one Hasse–Witt identity (Silverman, 2009, Chapter V), and can also be checked by the power sums

$$\sum_{w \in \mathbb{F}_p} w^k = \begin{cases} -1, & k > 0 \text{ and } p-1 \mid k, \\ 0, & \text{otherwise,} \end{cases} \qquad \text{in } \mathbb{F}_p. \tag{10}$$

For a cubic, only c_{p-1} survives in the sum of $f(w)^h$, and the projective curve has one point at infinity. For a quartic, $c_{2p-2} = f_4^h = (f_4/p)$ also survives; the infinity contribution is $1 + (f_4/p)$ and cancels this extra term. In both cases $\tau \equiv H \pmod p$.

Proposition 2 (Regular residue identity). Let $\lambda \neq 0$ be in any perfect extension of $\mathbb{F}_p$, with $f(1/\lambda) \neq 0$. Suppose $M \geq 1$, $p \nmid M$, and fix F_D with $\operatorname{div}(F_D) = MD$ as divisors. Then

$$M\left(S_f(\lambda) - H\right) = r\left(H\Psi_M(D) - \Psi_M(D)^p\right). \tag{11}$$

Proof. The residues cancel in $M\theta_\lambda + r\,dF_D/F_D$, so this differential is $c\omega$ for some scalar c. Evaluation at O gives $c = M + r\Psi_M(D)$. Apply Cartier, evaluate at O, and raise to the pth power to obtain

$$MS_f(\lambda) + r\Psi_M(D)^p = \left(M + r\Psi_M(D)\right)H. \tag{12}$$

This is (11). For the fixed divisor D, replacing F_D by F_D^k gives $\Psi_{kM}(D) = k\Psi_M(D)$ in characteristic p.

For a base-field argument, $\pi(Q_+) = Q_\chi$, so $\pi(D) = \chi D$ as divisors. Frobenius's characteristic equation annihilates its divisor class by $M_\chi = p + 1 - \chi\tau$. Since F_D^σ differs from F_D^χ by a constant, $\Psi_{M_\chi}(D)^p = \chi\Psi_{M_\chi}(D)$. Substitution into (11), using $M_\chi \equiv 1 - \chi H \pmod{p}$, proves (4) when $p \nmid M_\chi$. Changing the sign of v_0 negates both r and $\Psi_{M_\chi}(D)$.

Hasse's bound gives $|\tau| \leq 2\sqrt{p} < p - 1$ for $p \geq 7$. Thus $p \mid M_\chi$ occurs precisely when $\tau = \chi$, and then $M_\chi = p$. The class of D is nonzero: distinct points on a genus-one curve cannot have principal difference. It therefore has order p. The regular identity loses its scalar in this case. The bound excludes $p = 5$, where traces ± 4 are possible, from this particular degeneracy argument.

Direct exceptional normalization

Assume $\tau = \chi \in \{1, -1\}$ and identify $E = (v^2 = f(w), P_+)$, so $O = P_+$ and $\omega = dw/v$. Work over $k = \overline{\mathbb{F}}_p$. For a degree-zero divisor D_0 with p-torsion class, choose $\operatorname{div}\left(F_{D_0}\right) = pD_0$. Its logarithmic differential is holomorphic and defines

$$dF_{D_0}/F_{D_0} = \Psi_p\left([D_0]\right)\omega, \qquad \Psi_p : \operatorname{Pic}^0(E)[p](k) \to k. \tag{13}$$

A principal change $D_0 \mapsto D_0 + \operatorname{div}(g)$ changes the function by a constant times g^p and leaves its logarithmic differential unchanged. Multiplication of functions gives additivity. The map is injective by Serre's logarithmic-differential construction (Serre, 1958, Proposition 10). Since $\mathcal{C}\omega = \chi\omega$, its image lies in the one-dimensional $\mathbb{F}_p$-space

$$\mathcal{L}_\chi = \{b \in k : b^p = \chi b\}. \tag{14}$$

By the residue divisor of a differential we mean the formal sum of its residues at geometric points. When compared with an integer divisor, its coefficients are reduced in k, hence modulo p. In particular, logarithmic residues are integer multiplicities reduced modulo p.

Sublemma 3a (Separable pullback). Cartier commutes with pullback by a separable morphism of smooth curves. Indeed, in a separating coordinate x, a rational differential has a unique expression $\sum_{i=0}^{p-1} u_i^p x^i\,dx$, and Cartier selects $u_{p-1}\,dx$. A separable pullback keeps the coordinate separating, so applying the same expression after pullback proves the assertion.

For $\chi = 1$, the content of Lemma 3 is classical in substance. The proof follows the kernel differential and quasi-period construction in Voloch (1997): Θ_R in (18) corresponds to his differential $f_P\omega$, and (19) is its translation law. The Cartier invariance of logarithmic differentials is the setting of Gunji (1976) and Serre (1958). We retain a self-contained proof to fix the differential and translation conventions, treat $\chi = -1$ with its factor 2, and obtain the residue-divisor form needed by Theorem 1. The published discussion after Voloch's Proposition 3 prints a negative sign in its quasi-period/logarithmic-derivative formula. With $t_T(Q) = Q + T$ and divisor $p\left((T) - (O)\right)$, the divisor calculation in (17) gives the positive sign used here; the proof therefore determines the sign directly.

Lemma 3 (Exceptional Cartier defect). Suppose $\tau = \chi \in \{1, -1\}$. Let D_0 be a degree-zero integer divisor whose class is killed by p and whose coefficient conjugate is $\sigma(D_0) = \chi D_0$ as divisors. If θ_{D_0} is defined over $\mathbb{F}_{p^2}$, has only simple poles with residue divisor D_0, and satisfies $\sigma\left(\theta_{D_0}\right) = \chi\theta_{D_0}$, then

$$(\mathcal{C} - \mathrm{id})\,\theta_{D_0} = \Psi_p([D_0])\,\omega. \tag{15}$$

Equivalently, the Cartier-defect map and the logarithmic transfer are the same additive map from geometric p-torsion into $\mathcal{L}_\chi$; the normalization factor is exactly 1.

Proof. First consider $D_P = (P) - (O)$ for $\chi = 1$, or $D_P = (P) - (-P)$ for $\chi = -1$, where $P \neq O$ is geometric p-torsion. Let V be Verschiebung, identifying the Frobenius twist with E through its model over $\mathbb{F}_p$. The Frobenius relations and invariant differential give

$$\pi + V = [\chi], \qquad \pi V = [p], \qquad V^*\omega = \chi\omega, \qquad \mathcal{C}\omega = \chi\omega. \tag{16}$$

The curve is ordinary, so V is separable and étale of degree p. Its kernel $K = \ker V(k)$ is the cyclic group of geometric p-torsion, and every $T \in K$ automatically satisfies $\pi(T) = \chi T$. Thus $K = E\left(\mathbb{F}_p\right)$ if $\chi = 1$; if $\chi = -1$, it is the group $\{T : \pi(T) = -T\}$ corresponding to the rational points on the anomalous quadratic twist.

The divisor $\Delta = \sum_{T\in K}(T) - p(O)$ is principal: its group sum is O, because nonzero points pair as $T, -T$. It is Galois stable. The one-dimensional Riemann–Roch space $L(-\Delta)$ is defined over $\mathbb{F}_p$, so it has a nonzero base-field element s with $\mathrm{div}(s) = \Delta$. Put $\xi = \mathrm{dlog}\, s$ and $\psi(T) = \Psi_p([(T) - (O)])$. With translation $t_T(Q) = Q + T$, permutation of the kernel points gives

$$\mathrm{div}\,\frac{t_T^* s}{s} = p(O) - p(-T), \qquad t_T^*\xi - \xi = \psi(T)\omega. \tag{17}$$

The sign is positive because $[(O) - (-T)] = [(T) - (O)]$. Also $\mathcal{C}\xi = \xi$ and $\psi(T)^p = \chi\psi(T)$.

For $R \in E(k)$, put $Q = V(R)$. Equation (17) makes $t_{-R}^*\xi - \xi$ invariant under K. Division by $V^*\omega$ gives a K-invariant rational function, which descends through the separable quotient V. Hence there is a unique rational differential Θ_R on the target with

$$\begin{gathered} V^*\Theta_R = t_{-R}^*\xi - \xi = \mathrm{dlog}\,\tfrac{t_{-R}^* s}{s}, \\ \mathrm{div}\,\tfrac{t_{-R}^* s}{s} = V^*((Q) - (O)) - p((R) - (O)). \end{gathered} \tag{18}$$

As V is étale, Θ_R has only simple poles and residue divisor $(Q) - (O)$; the terms divisible by p have zero logarithmic residues. Sublemma 3a and injectivity of separable pullback give $\mathcal{C}\Theta_R = \Theta_R$. Translation and coefficient conjugation give

$$\Theta_{R+T} = \Theta_R - \chi\psi(T)\omega \quad (T \in K), \qquad \sigma(\Theta_R) = \Theta_{\pi(R)}. \tag{19}$$

Choose R with $V(R) = P$, so $\pi(R) = \chi R - P$. For $\chi = 1$, set $\alpha = \Theta_R$ and $\beta = \psi(P)$; for $\chi = -1$, set $\alpha = \Theta_R - \Theta_{-R}$ and $\beta = 2\psi(P)$. In each case α has residue divisor D_P and $\beta = \Psi_p([D_P])$. The factor 2 is required because $[(P) - (-P)] = 2P$. Equation (19) gives

$$\mathcal{C}\alpha = \alpha, \qquad \sigma(\alpha) = \chi\alpha + \chi\beta\omega, \qquad \beta^p = \chi\beta. \tag{20}$$

In the negative case, explicitly, $\sigma(\alpha) = \Theta_{-R-P} - \Theta_{R+P} = -\alpha - 2\psi(P)\omega$. A prescribed eigen differential θ_P differs from α by $c\omega$. Its eigencondition gives

$$c^p - \chi c = -\chi\beta, \qquad \chi c^{1/p} - c = \beta. \tag{21}$$

Thus $(\mathcal{C} - \mathrm{id})\,\theta_P = \beta\omega$.

To pass to D_0, represent its class by D_P, choosing half the class when $\chi = -1$; multiplication by 2 is invertible on the group of order p. The zero class is handled by $\mathrm{dlog}\, g$ directly. Choose a third-kind

differential η with residues D_P over $\mathbb{F}_{p^2}$ and replace it by $(\eta + \chi\sigma\eta)/2$ to satisfy the eigencondition. Conversely, the eigencondition implies descent to $\mathbb{F}_{p^2}$ because $\sigma^2\theta = \theta$. Write $D_0 - D_P = \operatorname{div}(g)$. Its eigencondition implies $g^\sigma = cg^\chi$ for some $c \in k^\times$, so $\operatorname{dlog} g$ is an eigen differential. The difference $\theta_{D_0} - \theta_P - \operatorname{dlog} g = b\omega$ has $b^p = \chi b$, hence is Cartier fixed. Cartier also fixes $\operatorname{dlog} g$. This proves (15).

This reduction shows that the defect depends only on the divisor class: changing an admissible differential adds a Cartier-fixed $b\omega$, and a principal eigen divisor contributes $\operatorname{dlog} g$, also Cartier fixed. The map is additive. Its image lies in $\mathcal{L}_\chi$, as does the injective transfer (13). The proof determines their comparison scalar as 1, including the factor of 2 in the negative eigencase. The kernel function and the preimage under V are proof constructions and are not computed by the evaluator.

Completion of Theorem 1. In the exceptional case put $\hat{\theta} = -\theta_\lambda/r$. It has residue divisor D and coefficient eigenvalue χ. Lemma 3 gives $(\mathcal{C} - \mathrm{id})\,\hat{\theta} = \Psi_p(D)\omega$. Since $r^{1/p} = \chi r$, evaluate at O to obtain

$$S_f(\lambda)^{1/p} = \chi - \chi r\Psi_p(D), \qquad \big(\chi r\Psi_p(D)\big)^p = \chi r\Psi_p(D). \tag{22}$$

Raising to the pth power proves (4) with $H = \chi$.

For concrete exceptional examples, direct expansion and the normalized Miller evaluator give the following values in $\mathbb{F}_{11}$, with $\lambda = 1$ and origin $(0, 1)$:

Table 2. Exceptional examples over $\mathbb{F}_{11}$.

$f(w)$	τ	$d = f(1)$	χ	H	$S_f(1)$	$r\Psi_p(D)$
$1 + w + 3w^3$	1	5	1	1	9	3
$1 + w + 2w^2 + 3w^3$	-1	7	-1	10	0	1

Remark on the high part. Put $\mathcal{G}_f(\mu) = \sum_{m \geq p} c_m \mu^m$. The Euler-criterion identity after (2) and Theorem 1 give

$$\mathcal{G}_f(\mu) = \chi - H + \chi r\Psi_{M_\chi}(D). \tag{23}$$

In the exceptional case $H = \chi$, this reduces to $\mathcal{G}_f(\mu) = \chi r\Psi_p(D)$. For a trace 1 short Weierstrass curve $y^2 = F(x)$, with origin at infinity and $\omega_0 = dx/(2y)$, the corresponding classical form is

$$F(x)^{(p-1)/2} = \mathcal{U}(x) + x^{p-1} + x^p\mathcal{V}(x), \qquad y\mathcal{V}(x) = \Psi_p\left([(P) - (O)]\right), \quad P = (x, y). \tag{24}$$

Here $\deg \mathcal{U} \leq p - 2$; the polynomial $\mathcal{V}$ is distinct from Verschiebung V. Its product with y is the Artin–Schreier descent function, and the right side is the logarithmic transfer (Deuring, 1941; Cassels, 1971; Voloch, 1990, n.d.; Semaev, 1998).

For completeness, fix a nonidentity rational point $P = (a, b)$. Since $E\left(\mathbb{F}_p\right)$ has odd order p, $b \neq 0$ and $F(a)^{(p-1)/2} = 1$. The differential $\vartheta_P = (y + b)\omega_0/(x - a)$ has simple poles with residue divisor $(P) - (O)$. Put $q(x) = \left(F(x)^{(p-1)/2} - 1\right)/(x - a)$. Polynomial division gives $\left[x^{p-1}\right] q = \mathcal{V}(a)$, and $\deg q < 2p - 1$. Applying Cartier to $dx/(x - a)$ and using $1/y = F(x)^{(p-1)/2}/y^p$ therefore yields

$$(\mathcal{C} - \mathrm{id})\vartheta_P = b\mathcal{V}(a)\omega_0. \tag{25}$$

Lemma 3, with $\chi = 1$ and differential ω_0, gives the displayed transfer identity. At O, both maps are defined to be zero.

Relation between the two exceptional signs

For a trace -1 curve, the quadratic twist has trace 1. On geometric p-torsion, the negative Frobenius eigenspace becomes the rational eigenspace of the twist, so the underlying logarithmic transfer is the twisted trace-1 construction. Transporting that map still requires the chosen differential scale and the specified residue divisor. In short coordinates the divisor $(P) - (-P)$ represents twice $(P) - (O)$, which accounts for the factor 2. Thus twisting explains the relation between the signs, while Lemma 3 records the normalization for the actual two-pole representative used in Theorem 1.

Computing the logarithmic derivative

Use the quartic-to-Weierstrass construction with the rational point as origin (Connell, 1999, Section 1.2); the following formulas specify the version and scale required here. Write $f = 1 + f_1 w + f_2 w^2 + f_3 w^3 + f_4 w^4$, with $f_4 = 0$ for a cubic, and set

$$x = \frac{2\,(v+1) + f_1 w}{w^2}, \qquad y = \frac{4\,(v+1) + 2\,(f_1 w + f_2 w^2) - f_1^2 w^2/2}{w^3}. \tag{26}$$

Substitution gives the generalized Weierstrass equation with

$$a_1 = f_1, \quad a_2 = f_2 - f_1^2/4, \quad a_3 = 2f_3, \quad a_4 = -4f_4, \quad a_6 = a_2 a_4. \tag{27}$$

Near P_+, direct expansion gives

$$\begin{aligned} x &= 4w^{-2} + 2f_1 w^{-1} + f_2 - f_1^2/4 + O(w), \\ y &= 8w^{-3} + 4f_1 w^{-2} + (4f_2 - f_1^2)\, w^{-1} + O(1). \end{aligned} \tag{28}$$

Thus P_+ maps to the identity at infinity. The points at infinity on the quartic map to affine points on this model; the finite points $Q_\pm$ also have affine images $R_\pm \neq O$, since $1/\lambda \neq 0$.

Put $b_2 = a_1^2 + 4a_2$, $b_4 = 2a_4 + a_1 a_3$, $b_6 = a_3^2 + 4a_6$, and use $X = x + b_2/12$, $Y = y + (a_1 x + a_3)/2$. The short equation is $Y^2 = X^3 + AX + B$, where

$$A = b_4/2 - b_2^2/48, \qquad B = b_6/4 - b_2 b_4/24 + b_2^3/864. \tag{29}$$

All denominators are units for $p \geq 7$. The invariant differentials are proportional; comparing their leading terms in (28) fixes

$$\omega = \kappa \frac{dX}{2Y}, \qquad \kappa = -2. \tag{30}$$

For a point R on the short model, use Miller functions and their local constants

$$\begin{aligned} \operatorname{div}\left(f_{m,R}\right) &= m(R) - ([m]R) - (m-1)\,(O), \\ \alpha_m(R) &= \left[z^0\right] \tfrac{\operatorname{dlog} f_{m,R}}{dz}, \quad z = -X/Y. \end{aligned} \tag{31}$$

This is a Laurent coefficient of df/f, not an analytic logarithm, and multiplication of a Miller function by a scalar does not change it. The divisor recurrence is standard (Miller, 2004; Galbraith, 2012, Section 26.3.1). A nonvertical chord or tangent of slope ρ and a vertical line have expansions

$$\begin{aligned} Y - \rho X - \nu &= -z^{-3}\left(1 + \rho z + O\left(z^3\right)\right), \\ X - c &= z^{-2}\left(1 - cz^2 + O\left(z^4\right)\right). \end{aligned} \tag{32}$$

The constant logarithmic derivative is therefore $+\rho$ for the nonvertical line and 0 for the vertical line. Binary doubling and addition update the accumulator by

$$\alpha \leftarrow 2\alpha + \rho, \qquad \alpha \leftarrow \alpha + \rho, \tag{33}$$

respectively. Vertical and identity steps contribute zero. Doubling the accumulator is still necessary when the current point is O.

Since $\left[M_\chi\right] R_+ = \left[M_\chi\right] R_-$, the quotient $f_{M_\chi,R_+}/f_{M_\chi,R_-}$ has divisor $M_\chi D$. Its pole terms at O cancel, and $dX/(2Y) = \left(1 + O\left(z^4\right)\right) dz$, so

$$\Psi_{M_\chi}(D) = \frac{\alpha_{M_\chi}(R_+) - \alpha_{M_\chi}(R_-)}{\kappa}. \tag{34}$$

Compute without finding a square root in the two-dimensional étale algebra $\mathcal{A}_d = \mathbb{F}_p[s]/(s^2 - d)$. Run the chain with $v_0 = s$ and write its constant as $\alpha = A_0 + B_0 s$. Algebra conjugation gives the other chain. Equations (4) and (34) become

$$r\Psi_{M_\chi}(D) = \frac{2B_0}{\kappa\lambda} = -\frac{B_0}{\lambda}, \qquad S_f(\lambda) = H + \chi\frac{B_0}{\lambda}. \tag{35}$$

For $\chi = -1$, the algebra is a field; for $\chi = 1$, it is a product of two fields. The latter requires branch tests componentwise. An element $u = A_0 + B_0 s$ with norm zero and $u \neq 0$ has $B_0 \neq 0$ and supplies the root $(A_0/B_0)^2 = d$. Split the saved state by $s \mapsto \pm A_0/B_0$ and resume the remaining binary steps in both fields. At addition, first handle an identity point; for equal abscissae use a vertical step when ordinates are opposite, and a tangent when the points coincide, treating an ordinate of zero as vertical. Otherwise use the chord. At doubling, an identity or ordinate-zero point is a zero-contribution step; otherwise use the tangent. A nonzero zero divisor in an equality test can also split the state before choosing these branches, so a tangent in one component and a vertical addition in the other are handled separately. Splitting can occur only once. This is the degree-two instance of dynamic evaluation, or the D5 principle (Della Dora et al., 1985).

Algorithm 1 (Differentiated Miller chain). The following pseudocode uses the short model and computes the constant in (31). Prepare the operation stream from the binary expansion of M, omitting its leading 1: each remaining bit contributes a doubling operation followed by an addition operation if the bit is 1. The stored operation index is the next uncommitted operation.

```
T := R; alpha := 0; j := 0
while j < length(operations):
    op := operations[j]
    Q := T if op is doubling, otherwise R
    attempt (Tnew, rho) := ADD(T, Q)
    if a nonzero zero divisor a + b*s is found:
        split saved R, T, alpha by s ↦ +a/b and -a/b
        resume both field components at this same j; return both constants
    anew := 2*alpha + rho if doubling, otherwise alpha + rho
    (T, alpha, j) := (Tnew, anew, j + 1)
return alpha, T
```

The subroutine ADD returns $(Q, 0)$ if $T = O$, and $(T, 0)$ if $Q = O$. For $T = (X_1, Y_1)$ and $Q = (X_2, Y_2)$, it returns $(O, 0)$ if $X_1 = X_2$ and $Y_1 + Y_2 = 0$; this includes doubling a point of order 2. If $T = Q$ otherwise, use the tangent slope $(3X_1^2 + A)/(2Y_1)$; if $X_1 \neq X_2$, use the chord slope $(Y_2 - Y_1)/(X_2 - X_1)$. Then set $X_3 = \rho^2 - X_1 - X_2$ and $Y_3 = \rho(X_1 - X_3) - Y_1$, and return $((X_3, Y_3), \rho)$. If $X_1 = X_2$ but the tangent and vertical tests disagree between components, $Y_1 - Y_2$ supplies the splitting witness. Any attempted inverse of a nonzero element of norm zero also supplies a witness. Its coefficient b is nonzero and $(a/b)^2 = d$. The split uses the saved state before the failed operation, so no accumulator update is repeated. In either component, identity and vertical steps still apply the doubling update when required. Finally use (34), or (35) before a split, to recover the coefficient sum.

For fixed f and p, cache the model, exact trace, H, and M_χ ($\chi \in \{1, -1\}$). Character computation and the affine Miller chain use $O(\log p)$ base-field multiplications and inversions. Resuming at most two components preserves this bound. The streaming query needs a constant number of working field elements and $O(\log p)$ control bits; an implementation retaining chain states may instead use $O(\log p)$ field elements.

Corollary 4 (Deterministic polynomial-time evaluation). Given $p \geq 7$, a squarefree cubic or quartic $f \in \mathbb{F}_p[w]$ with $f(0) \neq 0$, and an admissible λ, $S_f(\lambda)$ is computable deterministically in time polynomial in $\log p$. For N_{qry} queries on the prepared curve, classical arithmetic gives

$$T_{\mathrm{total}} = T_{\mathrm{Schoof}}(p) + O\left(N_{\mathrm{qry}}(\log p)^3\right), \qquad T_{\mathrm{Schoof}}(p) = O\left((\log p)^9\right). \tag{36}$$

The stated setup bound is the conservative bound of Schoof's original algorithm, not an optimal

point-counting bound. After setup, a query uses $O(\log p)$ field operations and at most $O(\log p)$ field elements plus $O(\log p)$ control bits. This query-space bound does not include the polynomial working space of point counting.

Proof. If $c_0 = f(0) \neq 1$, normalize $g = f/c_0$ and restore $S_f = c_0^h S_g$. Use the trace of $v^2 = g(w)$, which is a quadratic twist of the original curve when c_0 is nonsquare. Fixed-degree squarefreeness tests and the model change are polynomial in $\log p$. Schoof's deterministic point count gives the trace and annihilators (Schoof, 1985; see Schoof, 1995 for the survey). Theorem 1 includes exceptional traces. All query algebra has fixed dimension; a zero divisor supplies its own splitting root. Since $M_\chi = O(p)$, the chains have $O(\log p)$ length. Classical modular multiplication and inversion cost $O\left((\log p)^2\right)$ bit operations, giving (36). At $\lambda = 0$ use the trace residue of the normalized curve and restore c_0^h.

Practical trace preparation can use SEA or an automatic exact routine such as PARI/GP's `ellap`, with its available SEA data at large characteristic (PARI Group, n.d.). The exact trace can be cached once per curve and prime; a change of curve may require a new trace. These practical choices do not replace Schoof in the deterministic worst-case proof. The appendix uses specialized Cornacchia/Gauss preparation and does not benchmark a Schoof-based implementation of this corollary. For many queries, computing and sharing a full coefficient row can still be preferable.

Application to the original quarter-point sum

For an odd prime p, put $h = (p-1)/2$ and define the rational sequence

$$t_0 = 1, \qquad t_{i+1} = t_i \frac{h+2+i}{2i+2}, \qquad U_p(L) = \sum_{i=0}^{L-1} (h+1+i)\, t_i. \tag{37}$$

Reduction is well defined for $0 \le L \le h$, because the denominators use powers of two and factorials below p. Put $a_i = \binom{2i}{i} 8^{-i}$ and $B_L = \sum_{i<L} a_i$. Upper negation gives, in $\mathbb{F}_p$,

$$t_i = 2^{-i} \binom{h+1+i}{i} \equiv 2^{-i} (-1)^i \binom{-3/2}{i} = (2i+1) \binom{2i}{i} 8^{-i}. \tag{38}$$

The ratio $a_{i+1}/a_i = (2i+1)/(4(i+1))$ gives the telescoping certificate

$$\left((2i+1)^2 - 8\right) a_i = -4(i+1)(2i+7)\, a_{i+1} + 4i(2i+5)\, a_i. \tag{39}$$

Consequently,

$$U_p(L) \equiv 4B_L - 2L(2L+5)\, a_L \pmod{p}. \tag{40}$$

Both the prefix and the boundary coefficient are required for this weighted sum. Now take $p = 4n+1 \ge 13$, so $h = 2n$. Since $\binom{h}{i}(-1/2)^i = a_i$ in $\mathbb{F}_p$, B_n is a strict midpoint truncation of the binomial row of length $h+1$, although $n = (p-1)/4$ is a quarter-point relative to p. The full row sum is $(1/2)^h = (2/p)$.

Set $f(w) = 1 - w^4/2$. Its coefficient of w^{4i} in f^h is a_i, so $H = a_n$ and $S_f(1) = B_n + a_n$. Theorem 1 and (40) give

$$B_n = -\chi r \Psi_{M_\chi}(D), \qquad U_p(n) \equiv 4B_n + \frac{9}{4} a_n \pmod{p}, \tag{41}$$

where $\chi = (2/p)$ and $r = 1/v_0$, $v_0^2 = 1/2$. The curve has $j = 1728$ and rational nonzero two-torsion, so its trace is even and the exceptional traces ± 1 do not occur in this particular application.

The trace is classical complex-multiplication data. With $p = u^2 + s^2$ and the odd coordinate normalized by $u \equiv 1 \pmod 4$, Gauss's congruence is

$$\binom{2n}{n} \equiv 2u \pmod{p}, \qquad a_n \equiv 2u\, 8^{-n} \pmod{p}. \tag{42}$$

This convention agrees with Cosgrave and Dilcher (2010, Theorem 1). For $p \geq 17$, Hasse's bound $2\sqrt{p} < p/2$ makes the trace the representative of a_n in $(-p/2, p/2)$; the remaining prime $p = 13$ has trace 4, checked directly. Cornacchia preparation and exponentiation supply the boundary and trace for a Miller evaluation of the prefix. The benchmarked preparation includes a sequential nonresidue scan. Corollary 4 gives the deterministic polynomial-time alternative by computing the exact trace, using $a_n = H$, and evaluating $B_n = S_f(1) - H$. At $p = 97$, $n = 24$, $B_n = 33$, $a_n = 79$ and $U_p(n) = 43$.

The congruence literature for truncated central-binomial sums includes Sun (2011). The example here makes no separate priority claim for its binomial consequences. The elliptic-Fermat-quotient interpretation is established in the anomalous setting of Satoh and Araki (1998, 1999); applying that terminology to the regular $\Psi_{M_\chi}(D)$ in this example is only an analogy, since it retains a specified divisor and annihilator.

Rounded-third prefix and boundary

For primes $p \geq 11$ with $p \equiv 2 \pmod 3$, set $m = (p-2)/3$ and $L = m + 1 = (p+1)/3$. Retain the definitions of a_i, the strict prefix B_L and $U_p(L)$ from (37)–(40). Here the prefix is another direct application of Theorem 1: take $f(w) = 1 - w^3/2$ and $\lambda = 1$. The coefficients of f^h at w^{3i} are a_i, and $3m < p \leq 3L$. Thus $S_f(1) = B_L$. The coefficient H vanishes because $3 \nmid p - 1$; the exact trace is zero by the trace congruence and Hasse's bound. Since $f(1) = 1/2 \neq 0$, the argument is admissible, and the differentiated Miller query evaluates B_L in $O(\log p)$ field operations, without a point-counting search in this family.

The boundary a_L needed in (40) has a different arithmetic interpretation. Put $G = m! \bmod p$. Wilson reflection and the ratio of consecutive terms give

$$a_L = \frac{3}{2G^3}, \qquad G = \left(\frac{3}{2a_L}\right)^{(2p-1)/3} \quad \text{in } \mathbb{F}_p. \tag{43}$$

Indeed, $(2m)!L! = -1$ and $L! = LG$ give $(2m)! = -3G^{-1}$. Also $8^m = 1/2$, so $a_m = -6G^{-3}$, while $a_L/a_m = -1/4$. Cubing is bijective on $\mathbb{F}_p^\times$, yielding the inverse formula. Thus either a boundary backend or a modular-factorial backend supplies the other with $O(\log p)$ additional field operations.

The cubic above becomes $E : y^2 = x^3 + 1/4$ under $x = -w/2$, $y = v/2$. For the following cohomological statement use the point at infinity O as origin and fix $\omega = dx/y$; this differential has the opposite sign to dw/v under that coordinate change. Write $F = x^3 + 1/4$ and $F^h = \mathcal{T} + x^p\mathcal{S}$, with $\deg \mathcal{T} < p$. Let U be E with O removed, and W be E with $(0, 1/2)$ and $(0, -1/2)$ removed. A Čech–de Rham cocycle is (α_U, α_W, g) with $\alpha_U - \alpha_W = dg$, and a coboundary is $(du_U, du_W, u_U - u_W)$. Fix

$$e_1 = [(\omega, \omega, 0)], \qquad e_2 = \left[\left(x\omega, \frac{\omega}{2x^2}, \frac{2y}{x}\right)\right]. \tag{44}$$

Proposition 5 (Normalized rounded-third boundary). The classes (e_1, e_2) form a basis of $H^1_{\mathrm{dR}}(E/\mathbb{F}_p)$. With the Cartier convention of (8),

$$d(y\mathcal{S}) = a_L\omega, \qquad \mathcal{C}(\omega) = 0, \qquad \mathcal{C}(x\omega) = \mathcal{C}\left(\frac{\omega}{2x^2}\right) = 2a_L\omega. \tag{45}$$

Define $V_{\mathcal{C}}[(\alpha_U, \alpha_W, g)] = [(\mathcal{C}\alpha_U, \mathcal{C}\alpha_W, 0)]$ and $F_{\mathrm{abs}}[(\alpha_U, \alpha_W, g)] = [(0, 0, g^p)]$. These maps are linear over $\mathbb{F}_p$. With columns giving images of basis vectors,

$$[V_{\mathcal{C}}] = \begin{pmatrix} 0 & 2a_L \\ 0 & 0 \end{pmatrix}, \qquad [F_{\mathrm{abs}}] = \begin{pmatrix} 0 & -2a_L \\ 0 & 0 \end{pmatrix}. \tag{46}$$

Proof. For a polynomial P, put $\mathcal{D}(P) = FP' + F'P/2$, so $d(yP) = \mathcal{D}(P)\omega$. Since $h + 1/2 = 0$ in the field, $\mathcal{D}(\mathcal{T}) = -x^p\mathcal{D}(\mathcal{S})$. The expansion of F^h has exponents divisible by three, with $\deg \mathcal{T} = p - 2$,

so $\deg \mathcal{D}(\mathcal{T}) \le p$. Consequently $\mathcal{D}(\mathcal{S})$ is constant, equal to $[x^{p+1}]\, F^h/4 = a_L$. Cartier coefficient selection gives $[x^{p-1}]\, F^h = 0$, $[x^{p-2}]\, F^h = 2a_L$ and $[x^{p+1}]\, F^h = 4a_L$, proving (45).

The identity $d(2y/x) = (x - 1/(2x^2))\,\omega$ verifies the second cocycle. Its transition function has a simple pole at O. A Čech coboundary would require a function regular away from O with just that simple pole, excluded by genus-one Riemann–Roch. Together with the nonzero regular differential class, it therefore gives a basis. Applying Cartier proves the first matrix. For the second, write $(2y/x)^p = u_U - u_W$ with $u_U = 2y\mathcal{S}$ and $u_W = -2y\mathcal{T}/x^p$. These functions are regular on their respective opens, and $du_U = du_W = 2a_L\omega$. Subtracting their coboundary from $(0, 0, (2y/x)^p)$ gives $-2a_L e_1$. This proves (46). The companion extension supplies the regularity and basis arguments in full and states the Frobenius-twist conventions over perfect extensions (Oda, 1969).

The usual genus-one Cartier–Manin matrix on regular differentials is the single entry zero. The nonzero entry $2a_L = 3G^{-3}$ in (46) belongs to full de Rham cohomology in this specified basis. Although $a_L\omega$ is exact in the function field, it is nonzero in projective de Rham cohomology: its rational primitive has poles and cannot be made globally regular. This is a global regularity obstruction, not a prohibition on cancelling a pole at one point. Neither the zero trace nor this interpretation provides a fast boundary extractor, and no identification with a normalized Chowla–Selberg period is used.

Complete rounded-third evaluation

The boundary in (43) also supplies the adjacent cutoff $m = L - 1$. The consecutive-term ratio and (40) give

$$a_m = -4a_L, \qquad B_m = B_L + 4a_L, \tag{47}$$

$$U_p(L) = 4B_L - \frac{17}{3}G^{-3}, \qquad U_p(m) = 4B_L - \frac{16}{3}G^{-3}. \tag{48}$$

In particular, $U_p(L) - U_p(m) = -G^{-3}/3 \neq 0$, so the adjacent sums cannot both vanish. Here $G \neq 0$ because $m < p$. The boundary multiplier $2L(2L + 5) = 34/9$ vanishes only at $p = 17$ in this domain. Although the weighted sum then simplifies, the implementation still acquires and checks a_L and G separately.

The integrated single-prime routine validates p, obtains B_L from the unchanged differentiated Miller provider, acquires the signed boundary a_L by the selected backend, and assembles (40). Formula (43) recovers G when the backend supplies a_L directly; (48) then supplies the adjacent outputs. Apart from boundary acquisition, the complete routine requires $O(\log p)$ field operations, including modular powering. It returns the same tuple $(B_L, a_L, U_p(L), G)$ for every backend. The implemented public interface checks primality exactly in the supported domain $11 \le p < 2^{63}$, $p \equiv 5 \pmod 6$.

Certified Jacobi boundary acquisition

Write $k_\ell = (p-1)/\ell$ for a prime conductor $\ell \mid p - 1$. The bounded backend, called T3 in the code, supports primes $5 \le \ell \le 89$, with $k_\ell \ge 4$ required for forced calls. It reconstructs a Jacobi sum in $\mathbb{Z}[\zeta_\ell]$ by lattice reduction, accepts it only after the algebraic and normalization certificates, and evaluates its Galois conjugates to obtain grid-factorial relations. In particular, the norm relation $J\overline{J} = p$ is combined with the required root-of-unity normalization; a norm check alone is not the full certificate. Integer relation plans are checked on use. Two interval products connect the grid to G and to a rounded-third anchor. The interval lengths are approximately $k_\ell/3$ and $k_\ell/6$, or $k_\ell/3$ and $k_\ell/12$ when the denominator-12 anchor is available for $p \equiv 5 \pmod{12}$. The companion gives the exact residue-class formulas, relation plans, normalizations and certificates (Jordan, 2026a).

The denominator-12 anchor means an auxiliary factorial at $N = \lfloor (p-5)/12 \rfloor$ or $5N+2$, corresponding to rational fractions with denominator 12. It does not assert $12 \mid p - 1$; only the prime Jacobi conductor ℓ must divide $p - 1$. The saving lies in the anchor interval: the rounded-third relations replace the longer grid normalization by a shorter anchor tied to the same target. The signed half-factorial is not required, and neither class numbers nor GRH enter this integrated route. These

are bounded specializations of Jacobi and modular-factorial methods, not a new general complexity theorem. For fixed ℓ, the interval work retains square-root dependence on p up to logarithmic and multiplication factors. Lattice reconstruction is not proved to succeed on every input; exact acceptance and guaranteed reconstruction are different claims.

The automatic policy uses native FLINT factorial below 10^7. Above that threshold it ranks eligible conductors with $k_\ell \geq 10^4$ by a frozen cost model and attempts T3 when its prediction beats the nodal alternative. Certified reconstruction misses are recorded; automatic mode tries the remaining candidates and then falls back to the exact nodal backend. Forced T3 raises an error if reconstruction cannot be certified. The threshold is an implementation policy, not a universal crossover. The nodal backend is the companion's factorial-ratio anchor evaluated by FLINT block products; it uses no class number. Native factorial remains an explicit comparator and alternative. This fallback policy preserves the mathematical output but does not guarantee the fastest route on every machine.

Remainder forests for dense prime tables

A remainder forest shares ordered integer products and their reductions across many primes. Product trees combine recurrence intervals and prime moduli; remainders descend to the leaves to recover the required product at each prime's cutoff. Processing successive smaller trees, with a carry reduced modulo the unprocessed primes, controls the size of the retained arithmetic state. This is established accumulating remainder-forest machinery (Harvey & Sutherland, 2016), specialized here to an upper-triangular recurrence.

Over the rationals put $\Delta_j = 4^j j!$, $\widehat{B}_j = \Delta_j B_j$ and $\hat{a}_j = \Delta_j a_j$. Then

$$\mathsf{A}_j = \begin{pmatrix} 4(j+1) & 4(j+1) \\ 0 & 2j+1 \end{pmatrix}, \qquad \mathsf{A}_{j-1} \cdots \mathsf{A}_0 = \begin{pmatrix} \Delta_j & \widehat{B}_j \\ 0 & \hat{a}_j \end{pmatrix}. \tag{49}$$

This follows from $B_{j+1} = B_j + a_j$ and the ratio preceding (39); the empty product is the identity. At $j = L < p$, division by the nonzero Δ_L recovers B_L and a_L, while $G = \Delta_L / (4^L L)$ and (40) recover the other outputs. Thus the forest computes the complete tuple jointly and bypasses the Miller pass. Its gain is batch amortization, not an acceleration of an isolated Miller query. The companion provides ordered-product and carry invariants, bit-size and storage bounds, and reproduction commands (Jordan, 2026b, 2026a).

Dense prime schedules share enough recurrence work to benefit strongly. A sparse schedule still pays for products from index zero to its last requested cutoff, so independent calls can be preferable. The experiments compare these workloads separately. The integrated single-prime adapter does not automatically replace the workbench's existing many-prime dispatcher.

Supported evaluation modes

Table 3. Supported evaluation modes. The table distinguishes the geometric query from any additional boundary arithmetic.

Evaluation	Prefix or coefficient sum	Additional arithmetic
General admissible genus-one sum	Differentiated Miller query	Exact trace preparation
Original quarter-point weighted sum	Cartier–Miller prefix	Classical CM boundary and trace data
Rounded-third sum at one prime	Cartier–Miller prefix with known zero trace	Native factorial, nodal or certified Jacobi boundary
Rounded-third table across primes	Joint recurrence through a remainder forest	Shared prefix and boundary acquisition

For suitable individual primes the integrated Jacobi route improves complete evaluation, while dense tables favour the retained forest. Tables 5–7 give the matched measurements.

Scope and an independent point sum

At a branch argument $f(1/\lambda) = 0$, the two simple poles of θ_λ merge into a residue-free pole of the second kind. Removing exact differentials leaves a second-kind cohomology class whose Cartier image involves an additional polynomial-power coefficient, beyond the trace. Katz's note (n.d.) illustrates this distinction through Cartier on dx/y and $x\,dx/y$. This observation does not establish a fast branch evaluator. The current division-free streaming fallback takes $O(p)$ field operations.

If $f(0) = 0$, squarefreeness gives $f = wg$ with $g(0) \neq 0$, and the coefficient problem becomes $[w^h]\, g(w)^h/(1-\lambda w)$ rather than the normalized cutoff $p-1$. The origin and local coefficient selection used above therefore change; this case is not included in Theorem 1.

Proposition 2 already holds over perfect extension fields. For $q_k = p^k$ and $\lambda \in \mathbb{F}_{q_k}^\times$, put $e_k = d^{(q_k-1)/2} \in \{1,-1\}$ and define $\tau_0 = 2$, $\tau_1 = \tau$, $\tau_k = \tau\tau_{k-1} - p\tau_{k-2}$. The divisor class is annihilated by $M = q_k + 1 - e_k\tau_k$. If $p \nmid M$, (11) gives $S_f(\lambda) = H + (r/M)(H\Psi_M(D) - \Psi_M(D)^p)$, with a Miller chain of length $O(k\log p)$. Since $\tau_k \equiv \tau^k \pmod p$, the condition $p \mid M$ is equivalent to $e_k\tau^k \equiv 1 \pmod p$. The base-field simplification and equality with $T_f(1/\lambda)$ require separate arguments over extensions.

An elementary verification identity gives an independent $O(p)$ computation for admissible base-field arguments. With $\mu = 1/\lambda$ and $d = f(\mu)$,

$$S_f(\lambda) = \chi - \frac{\chi f'(\mu)}{2d\lambda} - \sum_{\substack{P\in E(\mathbb{F}_p)\ \text{affine}\\ w(P)\neq\mu}} \frac{1}{1-\lambda w(P)}. \tag{50}$$

To check it, write $F = f^h$ and $Q = (F - F(\mu))/(w-\mu)$. The integer number of affine points over w is congruent to $1 + F(w)$ modulo p. Summing after polynomial division, using (10) and $Q(\mu) = F'(\mu)$, yields the point sum $F(\mu) - T_f(\mu) + F'(\mu)/\lambda$. Since $F'(\mu) = -\chi f'(\mu)/(2d)$, this is (50). Points at infinity contribute zero if the summand is extended by its limiting value. This identity neither uses Miller functions nor proves their exceptional normalization by itself.

Define $\varepsilon_f(w) = \left(\frac{f(w)}{p}\right)$ first, with value zero at a root of f. The integer number of affine points above w is $1 + \varepsilon_f(w)$. Since $\sum_{w\neq\mu}(1-\lambda w)^{-1} = 0$ in $\mathbb{F}_p$, the same point-sum identity becomes

$$S_f(\lambda) = \chi - \frac{\chi f'(\mu)}{2d\lambda} + \mu \sum_{w\neq\mu} \frac{\varepsilon_f(w)}{w-\mu}. \tag{51}$$

The scalar $\chi = \varepsilon_f(\mu)$ is distinct from the character function ε_f. Substituting Theorem 1 in this display proves Corollary 1.1.

Higher genus

The residue cancellation behind Proposition 2 extends formally to a smooth curve of genus g: the holomorphic remainder after cancellation of the simple poles is written in a holomorphic basis as $\eta = \sum_{i=1}^g c_i\omega_i$, and $\mathcal{C}\eta = \sum_i c_i^{1/p}\mathcal{C}\omega_i$ involves the Cartier–Manin matrix. For a principal multiple of a divisor, logarithmic differentials and Jacobian transfer maps remain available (Rück, 1999). In genus one, however, every holomorphic differential is a scalar multiple of the fixed ω, so evaluation at one point determines the entire holomorphic correction. For $g > 1$ it supplies only one linear functional on a vector correction. Replacing H by a matrix alone therefore does not give the coefficient-sum formula or its exceptional normalization; these require further choices and arguments. No higher-genus evaluator is asserted here.

Computational verification

Retained code compares the generic evaluator with direct polynomial powers and independent point counts. An audit over primes $11 \le p \le 499$ tested 20,316 admissible queries on 906 sampled squarefree

cubics and quartics, including 368 exceptional queries and 2,512 split-algebra fallbacks, with no mismatches. The attribution revision reran this sample and additionally checked the high-part identity in the remark after Theorem 1 on every query; the character-sum form following (50) agreed on its 4,072 admissible queries with $p < 90$. Exhaustive compiled tests over $\mathbb{F}_5$ and $\mathbb{F}_7$ agreed with FLINT polynomial powers (Hart, 2010); the $p = 5$ checks are computational evidence outside the stated theorem.

A separate exhaustive check enumerated all 343 nonsingular short Weierstrass models $y^2 = x^3+Ax+B$ of trace 1 for $7 \le p < 48$. At all 11,730 nonidentity rational points, direct expansion of the high part agreed with the independent Miller slope accumulator relative to $dx/(2y)$. The identity point was checked separately on each model, and all 463,615 ordered pairs of rational points satisfied additivity. The supplementary $p = 5$ check covered two models and eight nonidentity points, outside the main theorem's prime range.

Separate retained direct-proof checks covered 17,940 nonzero torsion-point evaluations on 354 curves for $7 \le p \le 101$, and 568 kernel-translation checks on 32 curves for $7 \le p \le 31$. An independent point-sum verifier tested 4,357 queries on 384 curves for $7 \le p \le 103$, including 155 exceptional queries, with no mismatches. Source code, detailed logs and reproduction commands accompany the manuscript and the rounded-third companion extension. These finite checks do not replace proof review.

The earlier rounded-third companion has a separate validation record (Jordan, 2026b). Its anchor suite checked all 152 admissible primes below 2,000 and 595 reconstructed conductor values against direct character enumeration. Its native-FLINT comparison comprised 30 complete evaluations over five selected primes. Its million-bound standard-library batch had independent prefix checks, complete checks below 20,000 and 40 reproducible larger complete checks. These inherited records are distinct from the integration checks below; the unshared table below 100 million is not included in the independently verified evidence.

The integration candidate was internally hash-frozen before selecting fresh confirmation inputs; the commitment has no external timestamp. Development checked all 1,135 admissible primes below 20,000, with 3,405 complete method comparisons, 1,135 independent integer-binomial checks, 1,135 independently coded elliptic-prefix comparisons and 200 forced T3 checks. Three forest chunk counts agreed. Invalid domains and conductors, $p = 17$ and injected reconstruction misses were checked. The two relevant inherited test modules also passed, comprising 19 test methods.

Confirmation passed at 17 distinct primes, excluded from the recorded prior development, confirmation and benchmark ledgers, reaching $p = 7{,}000{,}000{,}009{,}553$. Every case compared complete native-factorial, nodal and automatic outputs and compared the Miller prefix with an independently coded elliptic provider. Applicable cases also used forced T3. The three approximately five-million inputs additionally received direct original weighted-recurrence, independent scaled-binomial-prefix and modular-binomial-product checks. At larger primes the original linear-time weighted recurrence was not run; agreement there uses independent prefix code and native-factorial boundary checks.

An exploratory forest survey produced all 39,264 admissible tuples below one million. The normalization and adjacent-sum residuals held throughout; neither $U_p(L)$ nor $U_p(m)$ vanished in that range. All 1,135 rows below 20,000 and 64 seeded rows above that bound received independent definition-level checks; the remaining rows received internal identity checks. The nonexistence of simultaneous zeros follows separately from (48). This survey is newly executed evidence on a previously studied range, not disjoint confirmation. The integration package records the selection rules, hashes, exact outputs and raw measurements (Jordan, 2026a).

Further questions

The identity concerns admissible genus-one inputs with a fixed coefficient cutoff. Projective arithmetic and a practical exact point-counting frontend could improve its implementation. Second-kind branch arguments, exceptional extension-field cases and other stopping families merit separate treatment. A

higher-modulus congruence would require new arguments: the characteristic-p Cartier proof does not establish evaluation modulo p^2.

For the rounded-third family, a uniform polylogarithmic single-prime extractor for the scalar in (46), equivalently the Gauss factorial in (43), remains open. The completed integration shows that better boundary acquisition can improve the full weighted-sum evaluation without changing the Miller prefix. A precise next question is whether guaranteed reconstruction at larger conductors can reduce interval work without its setup cost erasing the gain. The fixed-conductor implementation and de Rham interpretation do not settle that question. Further implementation work includes machine-calibrated workload selection and platform validation. No general computational lower bound is claimed.

Limitations

The proved scope is the admissible genus-one coefficient sum modulo p. The rounded-third boundary remains factorial-equivalent; its fixed-conductor implementation retains square-root interval dependence and certified acceptance does not guarantee successful lattice reconstruction. The benchmarks describe named implementations on one uncontrolled host with small samples; they establish neither universal crossovers nor superiority over all tools. The largest complete rounded-third check is approximately 43 bits. The regular quarter-point family has even trace and is not anomalous, so the trace-1 p-adic discrete-logarithm methods cannot be used as an independent cross-check without another comparison theorem. The native bridge was tested on Linux; Windows and future ABI compatibility remain unvalidated. Finite agreement and AI feedback are not specialist proof review. Priority for the particular residue-divisor and integration formulations requires a completed novelty audit.

AI assistance

Generative AI was used extensively in mathematical exploration, programming, computational checking and manuscript preparation under the author's direction. The reported computational checks arose within that research programme and the supplied AI feedback; they are not human peer review. The manuscript remains unrefereed by a human specialist.

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Appendix on computational benchmarks

Original quarter-point pilot

Table 4 presents a quarter-point pilot for seven primes $p \equiv 1 \pmod 4$, with $n = (p-1)/4$. Both implementations compute B_n, a_n and $U_p(n)$, including per-input preparation. The comparator is a standalone C++ adapter to the unchanged SageMath 10.8 Harvey `hypellfrob` interval-product kernel, which uses BGS-based recurrence techniques (Bostan et al., 2007; Harvey, 2007; SageMath Developers, n.d.). This is a benchmark of that recurrence task, not of Harvey's full Kedlaya algorithm.

Table 4. Complete quarter-point evaluation in milliseconds: Python Cartier–Miller versus the C++ Harvey/BGS recurrence adapter, including per-input preparation

Prime p	**Stopping index n**	**Cartier–Miller Python**	**Harvey BGS C++**
97	24	0.05	0.01
1,009	252	0.06	0.05
10,009	2,502	0.07	0.10

Prime p	**Stopping index n**	**Cartier–Miller Python**	**Harvey BGS C++**
100,049	25,012	0.09	0.54
1,000,033	250,008	0.10	1.78
10,000,121	2,500,030	0.11	4.44
100,000,037	25,000,009	0.29	16.19

The run used a shared hosted Linux environment reporting an AMD EPYC 9V74 processor, Python 3.12.14, GNU C++ 13.3.0 with `-O3`, and NTL 11.5.1 configured for one thread. Times are medians of 11 per-call timing-batch means, with alternating method order in warmed persistent processes. Batches targeted about 25 ms and were capped at 2,000 evaluations. Answers and per-call mathematical data were recomputed; library tables could remain warm. Validation, compilation, primality testing, process startup, communication and output writing were excluded. Host load and effective CPU frequency were not controlled.

All timed outputs agreed. Adapter validation additionally covered 10,809 stopping-point queries across 92 primes from 7 to 499, and 328 quarter-point comparisons for primes $p \equiv 1 \pmod 4$ from 13 to 4,999, with no mismatches. The timing run did not change these validation counts.

Cornacchia/Gauss preparation includes a nonresidue scan; this pilot does not implement Schoof setup or establish Corollary 4's worst-case bound. The direct characteristic-p proof did not change the evaluator. Small-input entries include call and loop overhead. Seven selected inputs, different implementation languages, an uncontrolled host and unmeasured memory limit general speed or scaling claims. Raw samples and reproduction commands accompany the manuscript.

Integrated rounded-third single-prime benchmarks

Tables 5–7 compare methods returning the identical complete tuple $(p, B_L, a_L, U_p(L), G)$. Each cell gives the median and observed minimum–maximum range of three samples with rotated method order on one host, using Python 3.12.14, python-flint 0.9.0 (FLINT 3.6.0), gmpy2 2.3.2 (GMP 6.3.0), Linux x86-64 and an AMD EPYC 9V74 processor. No concurrent number-theory job ran during timing; CPU frequency and scheduling were not controlled. These rounded-third measurements use a different protocol from the quarter-point pilot and must not be pooled with Table 4.

Each single-prime call rebuilds its Jacobi context and product trees and reruns Miller arithmetic; complete answers are not cached. Primality checks, registry and provider setup, conductor discovery, reconstruction certification, interval products, normalization, derived G and result collection are charged. Imports and prime selection are excluded, and reference comparisons occur after timing. A first call can load a prime-independent plan table or native-library handle. The native comparator calls FLINT's `n_factorial_fast_mod2_preinv` (FLINT developers, n.d.) for $m!$ and uses the same Miller prefix. The nodal comparator uses the same prefix with its factorial-ratio boundary.

Table 5. Complete rounded-third evaluation in milliseconds, shown as median [minimum, maximum] of three calls. Auto is the integrated automatic route; ℓ is its selected conductor.

Prime p	Auto route ℓ	Native + Miller	Nodal + Miller	Auto + Miller	Native / auto
70,000,283	T3 11	11.244 [11.013, 14.510]	9.716 [9.197, 11.064]	6.745 [6.105, 8.968]	1.67
70,000,481	T3 11	11.351 [11.004, 11.462]	9.547 [9.542, 11.833]	5.879 [5.386, 5.905]	1.93
70,000,613	T3 11	15.190 [14.775, 15.740]	12.261 [10.969, 12.432]	7.868 [6.523, 8.340]	1.93
7,000,000,013	T3 31	261.129 [254.152, 264.945]	220.053 [215.825, 251.297]	58.941 [56.682, 60.779]	4.43
7,000,000,277	nodal	256.361 [254.963, 259.401]	214.941 [208.206, 217.394]	214.746 [212.828, 215.143]	1.19
7,000,000,727	nodal	260.154 [255.525, 268.183]	214.452 [209.942, 215.550]	214.251 [212.405, 217.044]	1.21

Prime p	Auto route ℓ	Native + Miller	Nodal + Miller	Auto + Miller	Native / auto
7,000,000,769	nodal	263.522 [258.686, 265.291]	217.327 [212.077, 225.342]	225.696 [213.945, 226.674]	1.17
7,000,002,851	T3 23	259.000 [258.844, 273.559]	214.838 [213.861, 219.622]	63.885 [62.825, 65.143]	4.05
7,000,011,929	T3 13	266.407 [263.097, 283.127]	232.299 [218.851, 235.606]	77.799 [76.285, 85.781]	3.42
700,000,000,031	T3 71	5894.902 [5816.792, 6078.298]	4909.134 [4835.276, 4939.634]	1153.621 [1147.457, 1310.309]	5.11
700,000,001,633	T3 47	5833.161 [5831.019, 5857.249]	4760.048 [4749.239, 4880.084]	895.216 [868.507, 900.682]	6.52
700,000,004,549	T3 31	5751.378 [5718.851, 6103.010]	4784.071 [4749.360, 4909.331]	1014.534 [996.811, 1093.955]	5.67

All complete outputs agreed. At the nine inputs selecting T3, speedups over native factorial plus Miller ranged from 1.67 to 6.52. At $p = 700{,}000{,}001{,}633$, the complete median fell from 5.833161 s to 0.895216 s; the Miller prefix contributed approximately 0.48% of the new runtime. The no-small-factor inputs selected nodal fallback, so their gains over native are nodal gains, not T3 acceleration. Against direct nodal calls, automatic fallback has small selection overhead and timing variation.

Dense and sparse prime schedules

For Table 6 every method evaluates all admissible primes below the same exclusive bound. Sieve, arithmetic, collection and canonical tuple hashing are charged; disk I/O is excluded. Python and GMP forests use the same retained recurrence with different integer backends and acquire prefix and boundary jointly. Auto selects native throughout these small-prime ranges, so no dense-table gain is attributed to T3.

Table 6. Dense rounded-third tables, complete seconds as median [minimum, maximum] of three runs.

Exclusive bound	Tuples	Native + Miller	Auto + Miller	Python forest	GMP forest
2,000	152	0.120307 [0.120265, 0.122512]	0.122290 [0.120004, 0.126363]	0.001377 [0.001267, 0.001557]	0.001255 [0.001231, 0.001441]
20,000	1,135	1.273731 [1.272739, 1.344891]	1.336106 [1.278975, 1.344524]	0.015224 [0.013977, 0.016268]	0.011338 [0.010396, 0.017316]
100,000	4,805	6.677996 [6.650758, 6.870394]	6.652261 [6.646842, 6.750717]	0.157454 [0.155444, 0.161275]	0.057797 [0.056992, 0.057938]

At the 100,000 bound the GMP forest is approximately 115.5 times faster than independent native-factorial/Miller calls. This is the retained batch algorithm's amortization, not a speedup of the Miller pass. The forest need not be preferable for sparse schedules: it pays to construct recurrence products from index zero, rather than receiving the initial state near the first requested prime for free.

Table 7. One matched sparse schedule of sixteen primes from 1,000,211 to 2,501,171, complete runtime in seconds.

Method	Runtime in seconds
Native factorial + Miller	0.037191 [0.036596, 0.038203]
Automatic integration	0.037016 [0.036621, 0.039312]
Forced T3 $\ell = 5$	0.039155 [0.038894, 0.041857]
Python forest	8.171517 [7.951387, 8.624335]
GMP forest	0.689112 [0.679250, 0.817014]

The exact same sixteen-prime schedule and complete outputs were used throughout, with all output hashes agreeing. Independent native calls took 0.037191 s, versus 0.689112 s for the GMP forest, making the forest approximately 18.5 times slower here. Forced T3 at $\ell = 5$ did not improve this moderate-prime workload. Very large sparse-prime forest requests were outside the declared recurrence-cutoff cap of 2,000,000 and received no hypothetical timing.

The practical conclusion is workload-dependent: retain independent factorial and nodal alternatives, use certified Jacobi acquisition on suitable larger individual primes, and exploit a forest for dense tables. These experiments do not compare an optimized compiled forest, unmodified Tal reconstruction, the earlier project Jacobi backend, or complete PARI, SageMath and Magma systems. They establish improvements over the named matched comparators, not superiority over every available tool. The observed ranges are descriptive, not confidence intervals; in particular, modest fallback differences should not be interpreted as a robust universal speedup.

The reproducibility archive contains integrated.py, the original-definition oracles, the frozen selection rules, all 17 confirmation inputs, raw benchmark samples, the million-bound survey and preserved vendor proofs. README.md gives the commands for confirmation, benchmarks, survey and hash audit. The final file manifest and the earlier internal freeze have distinct roles: the freeze records the candidate before confirmation, whereas the manifest checks the delivered package. The standalone adapter integrates the existing boundary registry and complete-sum function; it does not constitute an updated graphical workbench release or a new many-prime dispatcher (Jordan, 2026a).